Model Question Paper-II

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Third Semester B.E. Degree Examination
Digital System Design using Verilog

TIME: 03 Hours | Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

			*Bloom's Taxonomy Level	Marks
		Module -1		
Q.01	a	What is combinational circuit? Design a combinational logic circuit with three input variables that will produce logic 1 output when more than one input variables are logic 1.	L2	06
	b	Convert the following Boolean function into canonical minterm and maxterm form in decimal format $R = f(a,b,c) = a + b(a+c) + bc$	L2	06
	c	Find all the prime implicants and essential prime implicants for the following function using k-map method. (i) $M = f(a,b,c,d) = \Sigma(1,5,7,8,9,10,11,13,15)$ (ii) $Y = f(a,b,c,d) = \pi(0,2,3,8,9,10,12,14)$	L3	08
		OR		
Q.02	a	. Simplify the following Boolean functions using K-map (i) $Y = f(a,b,c,d) = \pi(0,1,4,5,8,9,11) + d(2,10)$ (ii) $M = f(w,x,y,z) = \Sigma(0.1,2,4,5,6,8,9,12,13,14)$	L3	10
	b	Solve the following Boolean function by using QM minimization technique $P = f(w,x,y,z) = \Sigma(2,3,4,5,13,15) + d(8,9,10,11)$ Verify using K-map method	L3	10
		Module-2		
Q. 03	a	Implement the following functions using 3:8 decoder along with OR and/or NOR gates. In each case the gates should be selected so as to minimize their total number of inputs. (a) $f_1(X_2, X_1, X_0) = \Sigma\, m(0,2,4,6,7)$ and $f_2(X_2, X_1, X_0) = \Sigma\, m(1,3,5,6,7)$ (b) $f_1(X_2, X_1, X_0) = \Pi M(0,2,4,6,7)$ and $f_1(X_2, X_1, X_0) = \Pi M(1,3,7)$	L3	07
	b	Construct single decade decimal adder with necessary correction circuit design.	L2	08
	c	Design a one-bit comparator circuit.	L2	05
		OR		
Q.04	a	Construct the following function $S = f(a,b,c,d) = \Sigma(1,4,5,7,8,9,14,15)$ using (a)8:1 Mux and (b)16:1 Mux	L2	07
	b	Construct the functional table for 4 to 2 line priority encoder with a valid output, assigning highest priority to highest bit position or input with highest index and obtain the minimal sum expressions for the outputs.	L2	07
	c	Implement a Full adder using PAL	L2	06
		Module-3		
Q. 05	a	Explain the working of Master-Slave JK flip-flop with functional table and timing diagram	L2	08
	b	Derive the characteristic equations of SR and D Flipflops	L2	04
	c	Make use of negative edge triggered T -Flip Flops to describe the working	L2	08

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Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

			*Bloom's Taxonomy Level	Marks
		Module -1		
Q.01	a	Design a logic circuit that has 4 inputs, the output will only be high, when the majority of the inputs are high. Use k-map to simplify	L3	10
	b	Simplify the given Boolean function using Quine MC Cluskey find the prime and essential prime implicants and also verify with K-map $f(A,B,C,D)=\sum m(0,1,4,5,7,8,13,15)+d(2)$	L3	10
		OR		
Q.02	a	Simplify the following expression using K-map. Implement the simplified expression using basic gates only $f(a,b,c,d)=\Pi M(0,2,3,4,5,12,13)+dc(8,10)$	L3	10
	b	Define the following terms with example. Minterm, Maxterm	L1	04
	c	Place the following equation into proper canonical form $P = f(a,b,c) = ab'+bc$ $T = f(a,b,c) = (a+b)(b'+c)$	L3	06
		Module-2		
Q. 03	a	Design two bit magnitude comparator and write truth table, relevant expression and logic diagram.	L3	08
	b	Implement the following functions using 3:8 decoder $f1(a,b,c)=\sum m(1,3,5)$ $f2(a,b,c)=\sum m(0,1,6)$	L3	6
	c	Implement Y=ad+bc'+bd using 4:1 mux considering A and B as a select line.	L3	
		OR		
.04	a	Explain 4-bit carry look ahead adder with neat diagram and relevant expressions.	L2	
	b	Implement the following Boolean function using 8:1 multiplexer and 4:1 multiplexer $f(a,b,c,d) = \Sigma m(0,1,5,6,10,12,14,15)$.	L3	
		Module-3		
5	a	Explain the working of Master-Slave JK flip-flop with functional table and timing diagram.	L2	
	b	Explain Universal Shift Register with the help of logic diagram, mode control table.	L2	
		OR		
	a	Derive Characteristic equation for SR,T,D and JK flip-flop with the help of function table	L2	
	b	Design a Synchronous Mod-6 counter using SR flip-flop for the sequence 0-2-3-6-5-1.	L3	

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Third Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024

Digital System Design Using Verilog

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	a.	Develop a truth table of logic which takes two , 2 – bit binary numbers as its input and generate on output equal to 1, when the sum of the two numbers is odd.	7	L1	CO1
	b.	Convert the following Boolean function into : i) $f(abc) = (\bar{a} + b)(b + \bar{c})$ – Min term canonical form. ii) $f(xyz) = x + \bar{x}\,\bar{z}\,(y + \bar{z})$ – Max term canonical form.	7	L1	CO1
	c.	List the difference between Prime implicant and Essential prime implicant.	6	L1	CO1
		OR			
Q.2	a.	Simplify the given Boolean function using Quine Mc Cluskey Minimization Technique for the function $R = f(abcd) = \Sigma(0, 1, 2, 6, 7, 9, 10, 12) + dc(3, 5)$.	10	L1	CO1
	b.	Find the minimal sum and minimal product for the given function using K – map method for the function $R = f(abcd) = \Sigma m(0, 1, 3, 7, 8, 12) + dc(5, 10, 13, 14)$.	10	L1	CO1
		Module – 2			
Q.3	a.	List the difference between decoder and encoder and implement full adder using IC – 74138.	10	L3	CO2
	b.	What is Comparator? Design a 2 – bit digital comparator.	10	L3	CO2
		OR			
Q.4	a.	Realize the Boolean function $P = f(wxyz) = \Sigma(0, 1, 5, 6, 7, 10, 15)$ using i) 8 : 1 MUX ii) 4 : 1 MUX.	10	L2	CO2
	b.	With neat logic diagram, explain carry ahead adder.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain the working of master slave JK flip flop with help of Logic diagram , Function table , Logic symbol and Timing diagram.	10	L1	CO3
	b.	Obtain the characteristic equation for : i) SR flip - flop ii) J – K – flip - flop iii) D – flip - flop iv) T – flip - flop.	10	L2	CO3
		OR			